Math 1200 Lecture 7

Sep 24 2026

Topics: 14.6: Gradient, tangent plane, linear approximation Name:_____
Collaborators:_____

Group work 1 Calculate the directional derivative $D_{\vec{u}}f$ of $f(x,y) = \sin(xy) + 2x^2y$ of a unit vector in the direction $\vec{u} = \langle 3, 1\rangle$ at the point $(1, 0)$.

Group work 2 For each of the following functions $f(x, y)$

(a) Calculate the gradient ∇f

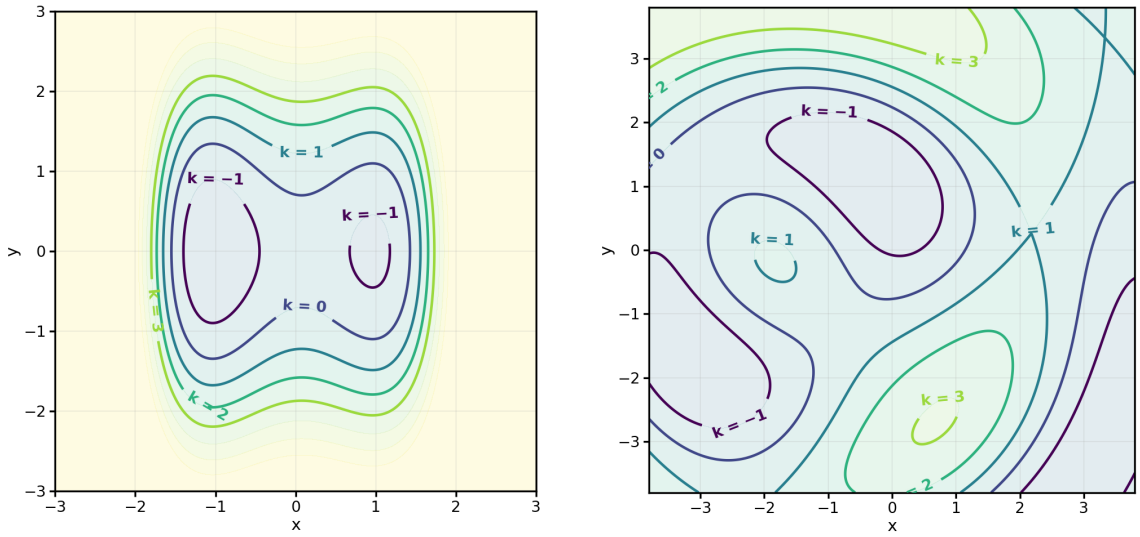
(b) Find the level curve $f(x, y) = \text{const}$ through the point $(1, 1)$

(c) Parametrize this curve using $\vec{r}(t)$

(d) Verify that $\vec{r}'(t)$ and ∇f are perpendicular at $(1, 1)$.

(1) $f(x, y) = 3x - y$ (2) $f(x, y) = 2x^2 + 2y^2$

Group work 3 Draw a few gradient vectors on the following contour plots:



Group work 4 (a) Consider the ellipsoid $x^2 + \frac{y^2}{2} + \frac{z^2}{3} = 1$. Find its tangent plane at the point $(x, y, z) = (1/2, 0, 3/2)$.

(b) Find a linear approximation to $f(x, y) = \sin(xy)$ at the point $(\pi/4, 1)$.

Scan to see a graph